

Noise-robust spoken digit detection

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Abstract

One of the fundamental computational challenges in speech recognition is preserving model fidelity in noisy environments. Decades of research have honed audio classification algorithms into very powerful tools, however the issue of noisy samples altering model accuracy persists. In this work, we investigated the role of acoustic feature engineering in improving the noise robustness of a fixed LSTM-based digit classifier trained on the Free Spoken Digit Dataset (FSDD). We evaluated a range of feature sets that we can capture from audio samples including Mel-Frequency Cepstral Coefficients (MFCCs), Linear Predictive Coding (LPC), Power-Normalized Cepstral Coefficients (PNCC), openSMILE feature configurations, and log-mel spectrograms processed through a Convolutional Neural Network. We trained our model exclusively on clean, professionally recorded audio samples, but evaluated it on clean audio samples as well as noisy samples captured in babble-noise-corrupted test conditions at SNRs of 5 dB and 10 dB.

We found that the combination of PNCC with delta, delta-delta, and CMVN achieves an average accuracy of 88.1% across all test conditions without augmentation and, additionally, 91.4% with augmentation, compared to 70.7% for the MFCC baseline, representing substantial gains particularly at the more challenging 5 dB SNR condition. These findings highlight that carefully designed feature pipelines, rather than simply larger feature sets or alternative classifiers, are the primary driver of noise robustness when training data is limited and the classifier architecture is held fixed.

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction

Digital speech processing (DSP) has burgeoned significantly in scope over the last decade. Innovations in computer science have allowed us to explore means of analyzing audio and speech patterns. Speech recognition tools are now ubiquitous and remarkably accurate thanks to developments in audio capturing hardware and DSP software. One of the most fundamental issues in the issue of speech recognition is the maintenance of accuracy in noisy environments. Creating accurate models that can classify audio samples is relatively easy when being both trained and tested on clean samples recorded either professionally or in silent environments. However, when deploying speech recognition products, there is no guarantee that audio from users will be of adequate quality for such a model. Translating these systems to real-world environments and maintaining performance is a significant software challenge.

Spoken digit recognition serves as a well-defined testbed for evaluating speech modeling techniques. Analogous to hand-

writing classification tasks in computer vision, this task requires a system to correctly classify isolated utterances of the digits zero through nine, and while it is simpler in scope than full continuous speech recognition, it requires us to build a model which maintains discriminative feature representations under adverse noise conditions. In this work, we train exclusively on clean speech data from the Free Spoken Digit Dataset (FSDD) [1], a publicly available dataset of 3,000 recordings across six speakers, and evaluate on both clean and babble-noise-corrupted test sets at signal-to-noise ratios of 5 dB and 10 dB. This training architecture allows us to evaluate models on both clean and noisy audio.

2. Background

Spoken digit recognition has been an active area of research in the field of DSP for several decades, serving as a benchmark task for evaluating acoustic modeling and feature extraction techniques. Early systems relied heavily on Dynamic Time Warping (DTW) paired with simple spectral features to align and compare spoken utterances of varying lengths. The introduction of Hidden Markov Models (HMMs) marked a significant shift, enabling sequence modeling from probability theory that better approximate speech signals. HMM-based systems dominated the field through the 1990s and 2000s, with MFCCs emerging as the standard acoustic feature due to their compact and perceptually motivated representation of the speech spectrum.

The advent of deep learning brought a new generation of sequence modeling architectures to speech recognition. Recurrent Neural Networks (RNNs), and in particular Long Short-Term Memory networks (LSTMs), proved especially well suited to the task by learning temporal dependencies across frame sequences without requiring explicit alignment. We know that RNN architectures could achieve competitive performance on speech recognition benchmarks, and subsequent work established LSTMs as a strong baseline for sequence classification tasks including digit recognition. More recently, Transformer-based architectures have pushed state-of-the-art performance further by replacing recurrence with attention mechanisms, though they typically require substantially more data and compute than LSTM models.

Nowadays, audio and speech processing tools and packages are ubiquitous and easy to access for any developer interested in DSP. Toolkits such as openSMILE [2] provide access to standardized feature sets drawn from the latest speech processing literature. The extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPSv02) is a compact configuration covering loudness, pitch, jitter, shimmer, harmonics-to-noise ratio, and low-order formant frequencies, while the ComParE_2016

set extends this with broader cepstral coverage including 26 RASTA-filtered auditory spectrum bands and 14 MFCCs. Although larger feature sets might be expected to capture more discriminative information, high-dimensional representations can introduce noise sensitivity and optimization difficulty when paired with a fixed classifier architecture, as reflected in our experimental results.

3. Project Description

3.1. Features

We used the openSMILE toolkit for acoustic feature extraction. openSMILE is an open-source framework designed to extract audio features in real time and supports both low-level descriptors (LLDs) and higher-level statistical functionals derived from these descriptors. The toolkit provides a wide range of acoustic features, including loudness, spectral descriptors, Mel-frequency cepstral coefficients (MFCC), pitch, formant frequencies, and voice-quality measures such as jitter, shimmer, and harmonic-to-noise ratio. In this work, we use predefined openSMILE feature configurations including eGeMAPSv02, a compact set of descriptors capturing loudness, spectral shape, MFCCs, pitch, and formants, and ComParE (2016), a larger feature set with broader cepstral and spectral coverage and additional voice-quality features. [2]

Power-Normalized Cepstral Coefficients (PNCC) were used as a noise-robust cepstral feature representation for speech signals. PNCC features are designed to improve robustness to background noise and channel variability compared to traditional MFCC features. The extraction pipeline begins with standard short-time spectral analysis of the speech signal, followed by frequency integration using a gammatone filterbank to approximate auditory frequency resolution. The method then applies several noise-suppression inspired processing steps, including medium-time power estimation, asymmetric noise suppression, and temporal smoothing, which help reduce the influence of background noise while preserving speech-relevant spectral structure. After these operations, power normalization and a power-law nonlinearity are applied to stabilize the dynamic range of the spectral features. Finally, a discrete cosine transform (DCT) is used to obtain the cepstral coefficients, followed by mean normalization. This sequence of processing stages emphasizes speech structure while suppressing environmental noise, making PNCC particularly effective when training and testing conditions differ in noise characteristics. [3]

Linear Predictive Coding (LPC) is a speech analysis technique that represents a signal using a linear prediction model. In this framework, the speech signal is modeled as the output of a system whose current sample can be approximated as a linear combination of a number of past samples. The resulting prediction coefficients describe the parameters of the system and provide a compact representation of the signal's spectral characteristics. In speech processing, this model is often interpreted as an excitation signal passed through a vocal tract filter, where the filter parameters determine the spectral structure of the speech signal. These coefficients therefore capture important information about the spectral envelope, including resonant characteristics associated with formants. Because LPC provides an efficient parametric representation of the speech spectrum, it has been widely applied in speech analysis, coding, synthesis, and recognition systems. However, the linear prediction model assumes relatively stable signal conditions, which can make LPC features sensitive to background noise and recording variability.

[4]

To incorporate temporal dynamics of speech, delta and delta-delta coefficients were appended to the static cepstral features. Delta features represent the first temporal derivative of a cepstral sequence and capture the rate of change of the spectral features across neighboring frames. In practice, the delta coefficient for a frame n is typically computed as the difference between cepstral values in neighboring frames, $D[n] = C[n+m] - C[n-m]$ where $C[n]$ denotes the cepstral feature at frame n and m represents the number of surrounding frames used in the regression window. Double-delta coefficients are obtained by applying the same temporal differencing operation to the delta features, thereby representing the second temporal derivative and capturing the acceleration of spectral changes over time. These dynamic features provide additional contextual information about speech evolution that is not present in static cepstral features alone, and they are therefore commonly appended to base features such as MFCC or PNCC in automatic speech recognition systems. The inclusion of delta and double-delta coefficients has been shown to significantly improve recognition accuracy by modeling temporal dependencies between frames in speech signals.[5]

Spectrogram-based representations are commonly used for audio classification because they provide a two-dimensional visualization of the frequency content of a signal over time. In this approach, the raw audio waveform is transformed into a Mel spectrogram, which captures the time-frequency structure of the signal in a format that resembles an image. Because the resulting representation is two-dimensional, it can be treated as an image and used as input to machine learning models designed for image processing, such as convolutional neural networks (CNNs). This representation allows models to learn discriminative patterns in the spectrogram that correspond to different sound classes, improving the ability to recognize audio events compared to using raw audio alone[6]

Cepstral Mean and Variance Normalization (CMVN). CMVN is a commonly used feature normalization technique in speech recognition that improves robustness to noise and channel variability. The method linearly transforms cepstral features so that each feature dimension has zero mean and unit variance, thereby reducing distortions introduced by environmental noise and recording conditions. By standardizing the statistical distribution of cepstral features, CMVN reduces variability caused by different acoustic environments and improves the consistency of the input features presented to the recognition model. This normalization step is typically applied after feature extraction and before the features are used for acoustic modeling in speech recognition systems. [7]

$$\hat{c}_d = \frac{c_d - \mu_d}{\sigma_d} \quad (1)$$

3.2. Supplementary Experiments

In addition to the official clean-train-only protocol, we ran a supplementary augmentation experiment using the best clean-train-only front end, PNCC+ Δ + Δ^2 +CMVN. During training only, Gaussian noise was added to the waveform with probability 0.6, with SNR sampled uniformly from 3 to 15 dB, before feature extraction. Validation and test data were unchanged. Since this modifies the training condition, we report it separately from the main comparison.

Furthermore, we implemented a Convolutional Neural Network (CNN) to test our hypothesis that classifying audio sam-

ples could be done at a high-level of accuracy using images of the sample spectrograms. Audio files were first converted to log-mel spectrograms using a 25 ms window with a 10 ms hop length across 64 mel filterbanks, then padded or truncated to a fixed width of 128 frames to ensure uniform input dimensions. These spectrograms were passed as single-channel 2D images into a three-layer CNN, where each layer applies a convolution, batch normalization, and ReLU activation followed by max pooling to progressively extract higher-level regional features. The final convolutional layer uses global average pooling to collapse the spatial dimensions into a 128-dimensional vector, which is then passed through a dropout layer and a fully connected output layer producing logit scores over the ten digit classes. Once the input layer was flattened and prepared for training, the rest of the network mirrored that of a traditional Artificial Neural Networks, trained on the same number of epochs as the original LSTM.

3.3. Implementation

All experiments were implemented in Python. We used ‘torchaudio’ for audio loading, ‘librosa’ for MFCC, LPC, and derivative features, ‘spafe’ for PNCC, and ‘opensmile’ for OpenSMILE descriptors. Experiments were run on a workstation with an NVIDIA RTX 4090 GPU.

To isolate the effect of the front end, we used a fixed classifier across all feature comparisons. The input feature sequence was processed by a two-layer bidirectional LSTM with hidden size 128 and dropout 0.2. Mean-pooled sequence outputs were then passed through a 256-to-32 linear layer, ReLU, dropout 0.3, and a final 10-way classifier.

All models were trained for 40 epochs with batch size 32 using Adam with learning rate 3×10^{-4} and cross-entropy loss. We applied gradient clipping with maximum norm 1.0 and used batch size 16 at evaluation. MFCC-based features used 13 coefficients with $n_{\text{fft}} = 256$, $\text{win_length} = 200$, and $\text{hop_length} = 80$, corresponding to a 25 ms window and 10 ms frame shift at 8 kHz. On the RTX 4090, non-augmented MFCC-, LPC-, and PNCC-based runs took about 1.6–2.0 minutes, OpenSMILE-based runs about 2.2–2.8 minutes, and the supplementary augmentation run about 15.9 minutes. Model selection used validation accuracy, and final evaluation was performed on clean, 5 dB babble, and 10 dB babble test sets.

3.4. Results

3.4.1. Main Results

We use accuracy as the primary metric because each test split is balanced across the 10 digits, with 30 utterances per class. Table 1 is best interpreted as a comparison of feature-set design choices under a fixed classifier and training protocol. Most systems are already strong on clean speech, so the main differences reflect how well each feature representation preserves discriminative structure under mismatch.

The best results come from normalized cepstral features with temporal derivatives. PNCC+ Δ + Δ^2 +CMVN is the strongest clean-train-only system with average accuracy 0.8811, followed by MFCC+ Δ + Δ^2 +CMVN at 0.8278. This gap suggests that once both front ends are given the same temporal and normalization machinery, PNCC provides the more robust base representation. The ablations also clarify where the gain comes from. For PNCC, performance rises sharply when CMVN is added, from 0.6511 to 0.8367, and then improves further with delta features. A similar, though smaller, trend appears

for MFCC, whose average accuracy improves from 0.7067 to 0.8278 after adding Δ , Δ^2 , and CMVN. Taken together, these results indicate that CMVN is the dominant factor, while temporal derivatives add complementary information about short-term speech dynamics.

The remaining feature families help clarify which design choices are less effective. LPC benefits slightly from CMVN and from combination with MFCC, but remains far below the strongest PNCC and MFCC systems, suggesting that prediction-based coefficients are less robust than cepstral features for this task. The log-mel+CNN system is more competitive: its average accuracy of 0.7278 exceeds the MFCC baseline, which indicates that local time-frequency structure is useful, but it still falls well short of the best normalized cepstral systems. OpenSMILE eGeMAPS shows a similar pattern. eGeMAPS alone is only modestly competitive, while eGeMAPS+MFCC improves further, suggesting that the handcrafted OpenSMILE descriptors carry some complementary information but are not strong enough to replace the cepstral features that dominate this task. By contrast, ComParE_2016 performs especially poorly, with average accuracy near 0.2, indicating that a much larger handcrafted descriptor set is a poor match for this small-data problem and likely introduces too many redundant or noise-sensitive cues.

Overall, Table 1 shows that robustness in this task is not improved simply by increasing feature dimensionality or descriptor complexity. Instead, the strongest results come from combining a robust cepstral front end with normalization and temporal dynamics, with PNCC+ Δ + Δ^2 +CMVN giving the best official clean-train-only performance.

Figure 1 compares the confusion matrices for the best clean-train-only system and its augmented counterpart. Under clean conditions, the model is nearly perfect, while the main errors emerge under babble noise. Digit 6 is the most difficult class overall. A likely reason is that the word “six” contains fricative energy at both the beginning and the end of the utterance, and these high-frequency, low-energy components are more easily masked by noise than the more stable voiced regions in other digits.

Table 1: Main clean-train-only comparison of feature sets on clean and noisy spoken-digit recognition.

Feature set	Clean	5 dB	10 dB	Avg.
PNCC+Δ+Δ^2+CMVN	0.9900	0.7633	0.8900	0.8811
PNCC+ Δ +CMVN	0.9767	0.6867	0.8633	0.8422
PNCC+CMVN	0.9767	0.6733	0.8600	0.8367
MFCC+ Δ + Δ^2 +CMVN	0.9800	0.6567	0.8467	0.8278
eGeMAPS+MFCC	0.9200	0.5667	0.7533	0.7467
Log-mel+CNN	0.8700	0.6100	0.7033	0.7278
LPC+MFCC	0.9800	0.5233	0.6667	0.7267
eGeMAPS	0.9167	0.5500	0.6800	0.7156
MFCC baseline	0.9800	0.4800	0.6600	0.7067
PNCC+ Δ	0.9233	0.4933	0.6700	0.6955
PNCC	0.9600	0.3867	0.6067	0.6511
LPC+CMVN	0.8800	0.4033	0.5500	0.6111
LPC	0.8467	0.3800	0.5033	0.5767
ComParE_2016	0.2100	0.2133	0.2167	0.2133
ComParE_2016+MFCC	0.1600	0.1967	0.2000	0.1856

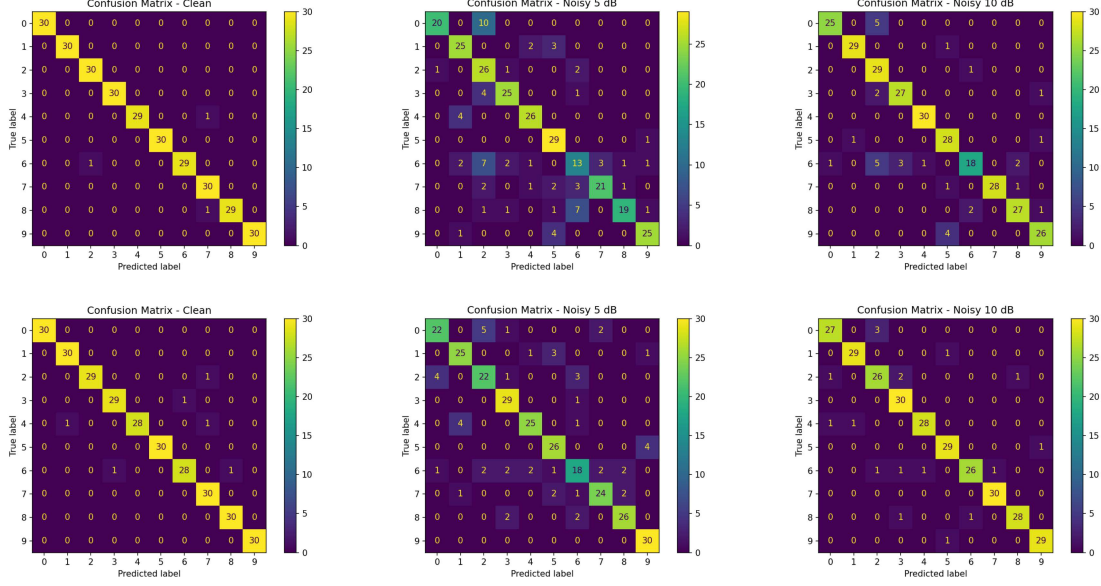


Figure 1: Confusion matrices for the best feature set, $PNCC+\Delta+\Delta^2+CMVN$. Top row: clean-train-only system. Bottom row: the same feature set with supplementary waveform-domain augmentation. Columns correspond to the clean, 5 dB, and 10 dB test sets.

Table 2: Supplementary aggregate metrics for the best feature set, $PNCC+\Delta+\Delta^2+CMVN$, with and without waveform augmentation.

Cond.	Split	Acc.	Prec.	Rec.	F1
Aug.	All	0.9144	0.9143	0.9144	0.9134
Aug.	Clean	0.9800	0.9805	0.9800	0.9799
Aug.	5 dB	0.8233	0.8214	0.8233	0.8204
Aug.	10 dB	0.9400	0.9411	0.9400	0.9398
No aug.	All	0.8811	0.8928	0.8811	0.8811
No aug.	Clean	0.9900	0.9905	0.9900	0.9901
No aug.	5 dB	0.7633	0.7899	0.7633	0.7645
No aug.	10 dB	0.8900	0.8980	0.8900	0.8888

3.4.2. Supplementary Results

As a supplementary experiment, we applied waveform-domain augmentation to the best clean-train-only front end. Table 2 shows that the same feature set remains strongest after noisy training, with average accuracy improving from 0.8811 to 0.9144. The largest gain appears at 5 dB, where accuracy rises from 0.7633 to 0.8233, while 10 dB accuracy improves from 0.8900 to 0.9400. The gains are consistent across macro precision, recall, and F1 as well, which is expected given the balanced evaluation splits.

We also examine per-digit averages across the clean, 5 dB, and 10 dB test sets. Table 3 shows that augmentation improves most digits, with the clearest gain for digit 6, whose average accuracy increases from 0.6667 to 0.8000. Digits 1 and 5 remain strong in both settings, while digit 6 remains the weakest class even after augmentation.

Figure 1 also shows the corresponding confusion matrices for the augmented system. Relative to the clean-train-only model, the augmented version reduces several off-diagonal errors in the noisy conditions. The improvement is visible across multiple digits rather than being concentrated in a single class,

Table 3: Supplementary per-digit average performance of the best feature set, $PNCC+\Delta+\Delta^2+CMVN$, with and without augmentation. Averages are computed across clean, 5 dB, and 10 dB test sets.

Digit	Acc. w/o	Acc. w/	F1 w/o	F1 w/
0	0.8333	0.8778	0.8924	0.8957
1	0.9333	0.9333	0.9244	0.9279
2	0.9444	0.8556	0.8168	0.8652
3	0.9111	0.9778	0.9158	0.9322
4	0.9444	0.9000	0.9444	0.9256
5	0.9667	0.9444	0.9052	0.9298
6	0.6667	0.8000	0.7177	0.8348
7	0.8778	0.9333	0.9037	0.9263
8	0.8333	0.9333	0.8760	0.9331
9	0.9000	0.9889	0.9145	0.9632

although digit 6 remains one of the weakest categories.

4. Summary and Discussion

Feature design was the main driver of robustness in this study. Under the official clean-train-only protocol, $PNCC+\Delta+\Delta^2+CMVN$ was the best system, substantially outperforming MFCC and all other front ends. The ablations show that this gain comes from the combination of a noise-robust cepstral base, temporal derivatives, and normalization: PNCC alone was not enough, but PNCC became highly effective once CMVN and temporal dynamics were added.

The weaker systems are also informative. LPC remained sensitive to noise, and OpenSMILE eGeMAPS did not match the strongest cepstral features. ComParE_2016 performed especially poorly, likely because its very high-dimensional descriptor set includes many redundant or noise-sensitive cues and is

poorly matched to this small-data task. In short, more features did not imply better robustness.

The supplementary augmentation experiment shows that the best feature set also benefits from noisy training. Waveform-domain augmentation improved accuracy, precision, recall, and F1, with the largest gains in the hardest 5 dB condition and for digit 6, the most noise-sensitive class.

Several extensions remain promising. LPCC could be tested as a more robust predictive feature than LPC. The fixed BiLSTM could be replaced by temporal CNNs, CRNNs, transformer-style encoders, or attention-based pooling. Finally, complementary feature combinations such as PNCC+MFCC or PNCC with selected OpenSMILE descriptors may provide additional gains if redundancy is controlled carefully.

5. Contribution

Xinwen Liu’s primary contribution was the development of the main Python script (entry point) for the corresponding feature-engineering experiments. This included implementation and experiments on the PNCC-based feature sets, OpenSMILE-based variants and data augmentation studies.

Karen Zhou’s main contribution was to test LPC, spectrogram, CNN, and delta/delta-delta features.

Alexander Sherzai’s main contribution was in testing LPC isolated, LPC in combination with other feature sets such as MFCC and CMVN and building the spectrogram-CNN pipeline.

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